

Bellman Conservation Law-based Crowd Flow Control via Passage Direction and Entrance-Rate Optimization in Metropolitan Tourism Hotspots With Capacity of 50,000 Pedestrians

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Abstract—Metropolitan tourism hotspots and major event venues face systemic crowd stampede risks due to unavoidable and stochastic dense-pedestrian flows, yet most existing crowd flow control models still relies mainly on empirical rules and post-hoc review rather than proactive, quantitative assessment. In such settings, managers’ main regulatory tools, passage directional configurations and time-phased entrance admission rates, still lack a quantifiable framework for jointly modeling and optimizing their coupled effects on global pedestrian flow. To address this academic issue, this paper develops a macroscopic crowd-control model based on the Bellman Conservation Law. Local admissible direction sets represent one-way, bidirectional, and closed passages, while anisotropic mobility tensors encode channel-guidance effects. Internal channel entrance-rate controls are added so that one policy specifies both passage directions and release rates. We then solve the mixed discrete-continuous problem with a Hierarchical Constrained Mixed-variable Black-box Optimization (HCMBO) algorithm. In a simplified scenic-platform scenario, the model reproduces multi-stage movement, channel guidance, and direction constraints. Under a weighted objective combining efficiency, safety exposure, load balance, waiting, and smoothness, HCMBO gives the lowest mean score within the tested budget. Its mean objective is about 3.9% lower than the Tree-structured Parzen estimator-based mixed-variable Bayesian optimization (TPE-Mixed BO) baseline and lower than the other tested baselines. These results can supply a large-scale qualitative basis to timely compare routing and entrance-release strategies in metropolitan tourism hotspots.

Index Terms—Crowd dynamics modeling, crowd flow control, mixed-variable black-box optimization.

I. INTRODUCTION

LARGE-SCALE crowd safety governance constitutes a critical component of urban governance and social security management. Crowd aggregation factors not only threaten individual life and health, but may also escalate into stampede incidents that undermine social order and public safety; the effectiveness of their mitigation directly reflects the modernization level of urban governance capacity and social security management systems [1], [2]. Tourism hotspots, public celebrations, commercial-district promotions, New Year events, and major sports gatherings increasingly concentrate pedestrian demand in core urban areas over short time windows [3], [4], [5]. These conditions strain on-site organization,

passage management, risk warning, and emergency dispatch. A central problem is therefore how to use modeling, simulation, and optimization to evaluate crowd-control strategies before an event, improve operational efficiency, and reduce local congestion risk [6], [7].

The difficulty of modeling large-scale pedestrian flow lies in the coupling between individual route choice and the collective spatiotemporal patterns it produces. Individuals with different desired speeds, destinations, and route preferences interact in limited space, so that microscopic decisions aggregate into macroscopic flow structure through nonlinear, multi-scale mechanisms [4], [8]. At the macroscopic level, these patterns appear as destination-oriented flow, passage convergence, bottleneck queues, local retention, exit competition, and directional responses to control rules [9], [10]. At the microscopic level, pedestrians respond to surrounding density, visible paths, obstacle boundaries, and guidance facilities [11]. However, reproducing every instantaneous pedestrian decision is not always the most relevant modeling target for crowd-management planning. The key issue is to choose a modeling resolution that can connect local behavioral responses with area-level congestion, load redistribution, and controllable operational measures.

Among urban crowd-gathering scenarios, open scenic areas, waterfront viewing platforms, and historic commercial districts are especially representative. Unlike enclosed facilities such as stadiums and subway stations with clear boundaries [12], [13], these areas combine open spaces, main tourist corridors, lateral connecting passages, and return flow lines. Visitors commonly follow a continuous behavioral chain of entry, touring along the main corridor, passage selection for leaving, and return movement [14]. Their movement is therefore shaped jointly by spatial geometry, control rules, and historical behavioral preferences [15]. For such open scenarios, sufficiently fine-grained and continuous individual trajectory data are often unavailable for calibrating detailed microscopic decision models. Recent calibration studies also show that microscopic pedestrian models can be sensitive to empirical travel-time distributions, bottleneck and exit-choice parameters, and measurement-error assumptions [16], [17], [18]. Field observations may be fragmented by camera coverage, occlusion, privacy restrictions, and changing temporary-control layouts. In contrast, managers usually need aggregate operational outputs: how regional density evolves over time, how loads are distributed across passages, how long queues or waiting masses accumulate near controlled

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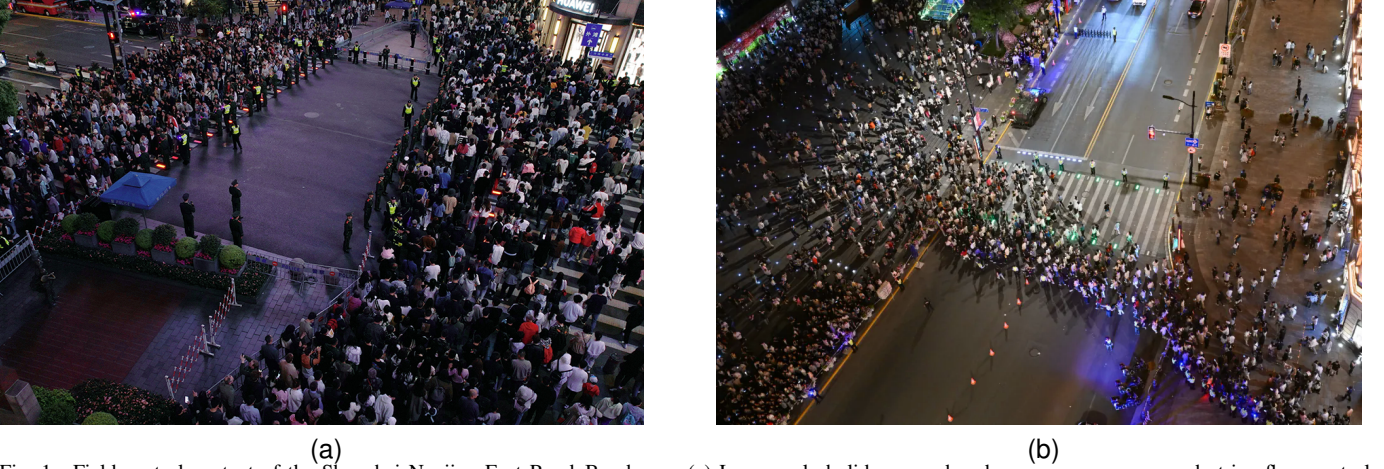


Fig. 1. Field-control context of the Shanghai Nanjing East Road–Bund area. (a) Large-scale holiday crowd under one-way passage pedestrian flow control on Nanjing East Road; image source: [19]. (b) Entrance-capacity pedestrian flow control link of Bund Passage No.5- Cheyi Square-Nanjing East Road; image source: [20].

entrances, and how much exposure occurs in high-density areas. These requirements motivate a macroscopic modeling perspective, in which pedestrian movement is represented through density evolution, route-stage subpopulations, passage fluxes, and entrance-release constraints rather than through the instantaneous decisions of every individual pedestrian. In the Shanghai Nanjing East Road–Bund area, for example, visitors approach the Bund from Nanjing East Road, ascend to the waterfront sightseeing platform through multiple stepped passages, move along the waterfront, and then leave through several southern passages before returning along lower streets. This layout can be abstracted as a “waterfront platform–multi-stepped passage” crowd-flow scenario. Its management difficulty comes from three coupled features: multiple passages simultaneously serve entry, exit, and diversion; transitions between the main corridor and lateral passages invalidate a single origin–destination description; and local congestion feeds back into global route choice, so changing one passage may redistribute load in adjacent areas. Fig. 1 illustrates two field-control contexts for this scenario: Fig. 1a shows a large holiday crowd under one-way passage organization, while Fig. 1b shows entrance-capacity control at a major visitor inflow location.

Existing security practice for large events usually combines system-dynamics reasoning, system coupling analysis, high-risk-zone identification, and historical crowd-behavior patterns to design operational controls [21], [22]. Typical measures include deploying police or staff at key locations, installing temporary barriers for separation and diversion, enforcing one-way passage rules, and releasing pedestrians in batches into core areas [23], [24]. During major holidays, for example, the Shanghai Bund area uses batch release, passage-direction regulation, south-entry–north-exit routing, and conflict-reduction measures [25]. Beijing Nanluoguxiang similarly applies one-way operation on the main street during peak periods, with tourists entering from the south and leaving from the north [26]. These cases share the same control logic: modify local direction and inflow conditions to reorganize pedestrian loads

at the area scale.

Despite these practices, high-pedestrian-flow management still depends heavily on manual experience and static contingency plans [27]. Many schemes do not quantitatively describe how passage-direction configurations, entrance release rates, and stage-dependent route preferences shape density evolution. Monitoring based on single-point counts or local density is also insufficient for the coupled process linking passage direction rules, entrance metering, local optimal directions, collective density evolution, and exit-load distribution [28]. In multi-entrance, multi-passage, and multi-stage scenic areas, neglecting route preference and behavioral phase transitions can misjudge passage load and high-risk locations. Moreover, many simulation-optimization studies treat the crowd simulator as an opaque evaluator, rather than exploiting structural information from passage geometry, one-way rules, internal entrance capacities, and multi-route subpopulations [29], [30]. This limits search efficiency and makes it difficult to separate whether a candidate strategy works because the underlying control mechanism behaves correctly or because a fixed aggregate score happens to improve.

Therefore, this paper proposes a macroscopic crowd modeling and control optimization framework for the “Bund sightseeing platform–multi-stepped passage” open scenic area scenario. The framework connects multi-stage pedestrian movement, passage direction management, geometric guidance, and entrance release control in one simulation-and-optimization workflow. It compares candidate strategies under limited simulation budgets, rather than relying only on empirical rules or post-event assessment.

Our main contributions are fourfold:

- 1) We formulate a multi-stage, multi-route macroscopic model for the open scenic-area class represented by the Bund sightseeing platform and its connected passages. Unlike single-origin–destination or single-potential Hughes-type formulations, the model lets entry, sightseeing, exit, return movement, and route redistribution arise through shared congestion.

- 2) We develop a Bellman–conservation-law operator that combines hard direction rules, anisotropic mobility guidance, and internal entrance-rate metering. This unified macroscopic operator represents one-way, two-way, and closed channels, channel pre-alignment, open-but-metered entrances, and upstream waiting under mass-consistent dynamics.
- 3) We establish an evaluation protocol that separates behavioral-mechanism diagnostics from strategy-level management objectives. The protocol first checks whether direction constraints and geometric guidance behave as intended, and then compares policies by efficiency, high-density exposure, load balance, entrance waiting, and control smoothness.
- 4) We design a hierarchical constrained mixed-variable black-box optimization framework (HCMBO) for joint direction and entrance-rate control. The method exploits the direction–capacity structure and uses unified high-fidelity rechecking to compare candidate policies under the tested budget.

II. RELATED WORK

A. Massive Crowd Management and Control

Crowd management studies in urban centers have mainly used boundary regulation, facility layout, and decision-making models [31], [32]. Zhong et al. [33] designed boundary control strategies for urban traffic-flow networks and proved convergence to uncongested states under disturbances through Lyapunov analysis [34]. Wadoo and Kachroo [35] used advection and diffusion control to prevent blocking and shock waves in one-dimensional evacuation. Qin [36] proposed a unit sliding-mode controller based on integral barrier Lyapunov functions for multi-directional disturbance propagation. Zhu et al. [37] combined policy learning and neural networks to adjust inflow rates and free-flow speeds in heterogeneous corridors.

Infrastructure-based regulation has also been studied. Obstacle placement can change outflow, velocity, and local density in merging areas [27], [38]. Yang et al. [39], [40] used Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks and later included entrance limiters, gates, and escalator directions. Carmona and Paricio-Garcia [41] proposed dynamic exit-choice recommendations using multinomial Logit models, but the approach depends on discrete speed instructions (e.g., “stop,” “walk,” “run”) transmitted through electronic wristbands [42], which are hard to deploy in open urban areas.

Other studies use optimization and reinforcement learning. Liao et al. [43] formulated fence layout as subset selection. Subsequent work [30] used long short-term memory (LSTM) features and Proximal Policy Optimization (PPO) [44] for real-time entrance-flow regulation. Differential Evolution (DE) [45] has been used to optimize guardrail layout and flow guidance in scenarios such as Chengdu East Railway Station, with average travel time and crowd pressure as objectives [46].

B. Macroscopic Crowd Modeling and Current Discrepancy Analysis

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency and analytical tractability compared to microscopic approaches. Henderson [47] first applied fluid dynamics to pedestrian traffic in the early 1970s, comparing crowd motion to the behavior of gas or fluid particles.

Hughes [48] pioneered a first-order continuum theory for pedestrian flow, in which walking speed depends on local density through a fundamental diagram, and movement direction follows the steepest-descent path of a potential function representing the remaining travel time to exits. The model links congestion and route choice through this potential. Ling et al. [49] generalized the Hughes formulation and solved it with a weighted essentially non-oscillatory (WENO) scheme for the conservation law and a fast-sweeping method for the eikonal equation.

Several extensions modify the density, perception, and constraint terms in macroscopic models. Colombo et al. [50] proposed nonlocal flux models that replace pointwise density with a neighborhood-averaged quantity in the velocity function to account for anticipatory behavior. Later work included anisotropic perception fields and wall or obstacle boundary effects, together with high-resolution schemes and well-posedness results for architectural domains [51]. Maury et al. [52] introduced a projection model for the hard incompressibility constraint, formulating the actual velocity as a least-squares projection of the desired velocity onto the set of feasible non-overlapping configurations. The same idea was later recast as a Wasserstein gradient flow with density constraints [53], using the Jordan–Kinderlehrer–Otto (JKO) scheme [54] to keep density below its upper bound.

To overcome non-physical behaviors such as supersonic information propagation that may arise in first-order models, Aw and Rascle [55] introduced a second-order model based on a generalized momentum conservation law that preserves anisotropy. The Aw–Rascle (AR) framework was subsequently adapted to pedestrian flows [56], where the congestion pressure term successfully reproduces characteristic phenomena including stop-and-go waves and spontaneous lane formation.

Beyond the classical Hughes-type framework, extensions have incorporated control and guidance mechanisms relevant to crowd management. Anisotropic formulations using mobility tensors have been developed to represent channel geometries and directional preferences [57], [58], allowing for continuous geometric guidance without explicit prohibition. Constrained Hamilton–Jacobi–Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors, with the Bellman optimality principle providing a natural framework for unifying anisotropic geometric guidance with discrete directional constraints.

Despite these advances, three gaps remain for large open scenic areas. First, Hughes-type single-potential models are destination-oriented and describe one population evacuating toward fixed exits [48], [49]; they cannot represent the multi-stage chain of entering, touring, leaving, and returning, in

which subpopulations with stage-dependent goals share the same congested space. Second, most macroscopic extensions refine the description of flow [50], [52], [56], whereas the operational levers used in practice, switching passages among one-way, two-way, and closed states and metering entrance release, are treated as fixed boundary conditions rather than as a joint discrete–continuous decision space [39], [30]. Third, local density or single-point counts are insufficient for such coupled systems [28]; strategy assessment requires area-level indicators linking regional density, passage loads, entrance queues, and exit allocation, which few macroscopic studies provide. These gaps motivate the multi-stage operator, the joint direction–capacity control space, and the evaluation protocol of this paper.

III. METHODOLOGY

A. Problem Definition and Overall Framework

We address a multi-channel crowd management problem in open scenic areas. The proposed method combines a Bellman–conservation-law crowd dynamics model [48], [49] with a joint direction–capacity control formulation. The controllable management variable is defined as

$$z = (s, q) \in \mathcal{Z}, \quad (1)$$

where s denotes the discrete direction states of the channels, and q denotes the time-varying internal entrance capacity profile. The behavioral preference parameter $\hat{p}$ and the geometric guidance parameter η_0 are treated as fixed model inputs. For a given control variable z , the lower-level simulator returns density fields, velocity fields, entrance waiting, and realized channel throughputs. The upper-level optimizer then evaluates efficiency, safety, load balance, waiting, and smoothness metrics, and searches for a better control policy under a limited simulation budget. The overall method framework is shown in Fig. 2.

We model direction rules and entrance capacities as coupled controls. The direction configuration s changes the local feasible direction set, whereas the entrance capacity q changes the realized admitted flux at channel entrances. Together, they affect channel loading, high-density exposure, and upstream waiting. We therefore formulate crowd management as a mixed-variable simulation-based optimization problem, rather than solving direction selection and capacity assignment as two independent subproblems [30], [59], [60].

B. Bellman–Conservation-Law Crowd Dynamics

1) *Conservation Law and Speed Function*: We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walking area. Let $\Omega_w \subset \Omega$ denote the walkable region, $\rho(x, t)$ the total crowd density, and $\mathbf{v}(x, t)$ the local mean velocity. Mass conservation is governed by

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (2)$$

The speed magnitude is determined by local congestion. We adopt the Greenshields-type speed function

$$f(\rho) = v_{\max} \left(1 - \frac{\rho}{\rho_{\max}} \right)_+, \quad (3)$$

where $v_{\max}$ is the free-flow speed, $\rho_{\max}$ is the maximum admissible density, and $(\cdot)_+$ denotes nonnegative truncation. This function captures the feedback that pedestrians move close to free-flow speed in low-density regions and slow down as density approaches congestion.

The classical Hughes model assumes that all movement directions are feasible and isotropic [48]. This assumption is insufficient for crowd management scenarios involving one-way rules, temporary closures, entrance metering, and channel-axis alignment. The following subsections introduce admissible direction sets, anisotropic mobility tensors, and internal entrance capacity control into the Bellman–conservation-law framework.

2) *Multi-Stage Multi-Route Density Decomposition*: Visitors in open scenic areas usually experience several behavioral stages, such as entry, touring, exit, and return. To represent this behavior chain, the total density is decomposed into stage–route subpopulations:

$$\rho(x, t) = \sum_{\sigma=1}^S \sum_{r=1}^{R_\sigma} \rho_{\sigma,r}(x, t), \quad (4)$$

where σ denotes the behavioral stage and r denotes the route type within that stage. All subpopulations share the speed function $f(\rho)$ determined by the total density, but they have different target regions, potential functions, and velocity directions.

For subpopulation (σ, r) , the density evolves according to

$$\frac{\partial \rho_{\sigma,r}}{\partial t} + \nabla \cdot (\rho_{\sigma,r} \mathbf{v}_{\sigma,r}) = Q_{\sigma,r}^{\text{in}} - Q_{\sigma,r}^{\text{out}}. \quad (5)$$

If pedestrians in subpopulation (σ, r) complete the current stage in target region $G_{\sigma,r}$ and switch to next-stage route k with probability $\hat{p}_{(\sigma,r) \rightarrow k}$, the transition term is

$$Q_{(\sigma,r) \rightarrow (\sigma+1,k)}(x, t) = \hat{p}_{(\sigma,r) \rightarrow k} \kappa_{\sigma,r} \chi_{G_{\sigma,r}}(x) \rho_{\sigma,r}(x, t), \quad (6)$$

where $\kappa_{\sigma,r}$ is the stage-switching rate, with dimension of inverse time, and $\chi_{G_{\sigma,r}}$ is the indicator function of the completion region. The parameter $\hat{p}$ represents historically inferred or scenario-given route preferences and is fixed in the optimization problem [61], [14], [15].

C. Direction Constraints and Geometric Guidance

1) *Direction Configuration Variable*: Let the direction state of channel c be

$$s_c \in \{+1, -1, 0, \emptyset\}, \quad (7)$$

where $+1$, -1 , 0 , and $\emptyset$ denote reference forward one-way, reference reverse one-way, bidirectional, and closed states, respectively. If $\tau_c(x)$ is the reference tangent direction of channel c , the local admissible direction set in the channel region is defined as

$$U_c(x; s_c) = \begin{cases} \{\tau_c(x)\}, & s_c = +1, \\ \{-\tau_c(x)\}, & s_c = -1, \\ \{\tau_c(x), -\tau_c(x)\}, & s_c = 0, \\ \emptyset, & s_c = \emptyset, \end{cases} \quad x \in \Omega_c. \quad (8)$$

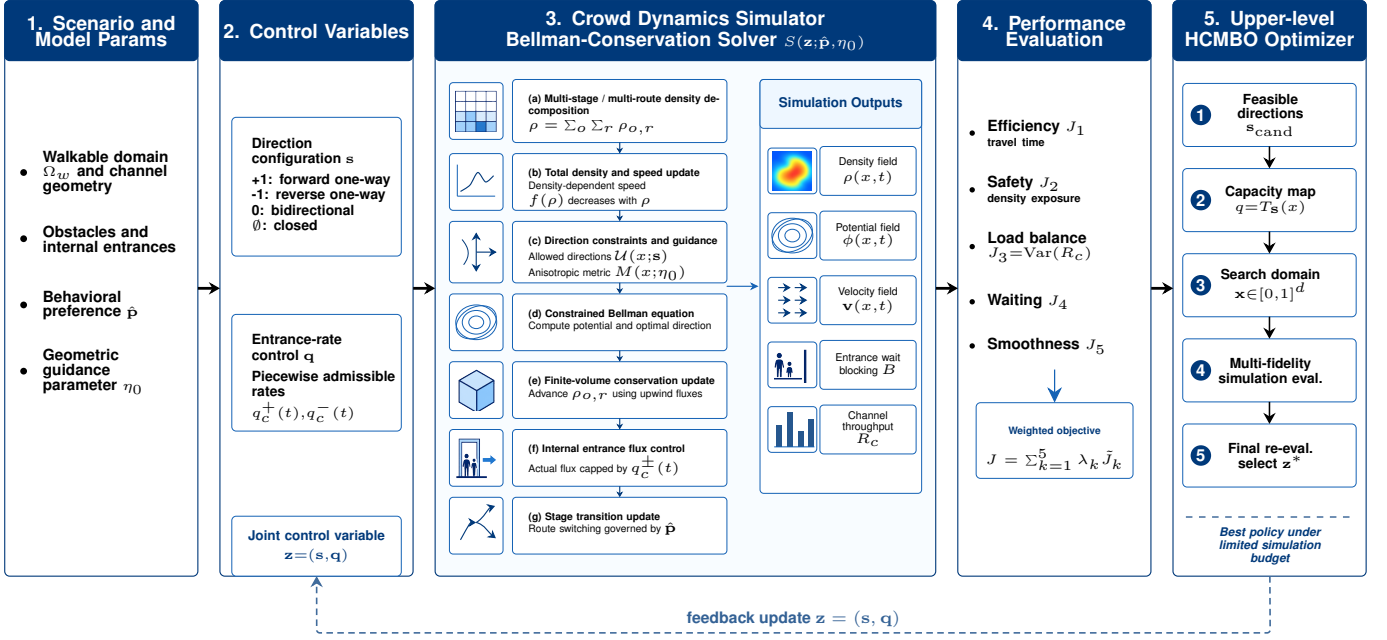


Fig. 2. Method framework for joint direction and entrance-rate optimization. Fixed scenario inputs, behavioral preferences, and geometric guidance parameters define the lower-level Bellman–conservation-law crowd simulator. The controllable direction configuration and entrance-rate profile are jointly evaluated by efficiency, safety, load-balance, waiting, and smoothness objectives, and are updated by the upper-level HCMBO optimizer.

Thus, s_c changes the local feasible direction set. A closed channel has no admissible direction, a one-way channel retains only one reference tangent, and a bidirectional channel retains both opposite tangential directions.

2) *Anisotropic Mobility Tensor*: The admissible set $U(x; s)$ represents hard feasibility, but it does not capture the soft geometric guidance induced by narrow channels. Pedestrians approaching a channel often pre-align with the channel axis before entering it. To represent this effect, an anisotropic mobility tensor $M(x; \eta_0)$ is introduced [51], [57], [58]. Equivalently, it is the inverse of a local travel-cost metric. For channel c , let τ_c and n_c denote its tangent and normal directions. The mobility tensor is defined as

$$M_c(x; \eta_0) = \beta_c (\eta_0 \tau_c \tau_c^\top + n_c n_c^\top), \quad \eta_0 \geq 1, \quad (9)$$

where β_c is a channel weight and η_0 controls the mobility preference strength along the channel axis. Larger η_0 increases tangential mobility, equivalently reducing the effective cost of tangential motion and promoting upstream pre-alignment. The parameter η_0 is fixed and is not optimized as a control variable.

3) *Constrained Bellman Equation*: Given $U(x; s)$ and $M(x; \eta_0)$, the potential function $\phi(x, t)$ represents the minimum remaining cost to the corresponding target region. The discrete Bellman update is

$$\phi(x) = \min_{\mathbf{u} \in U(x; s)} \left[\phi(x + \Delta x \mathbf{u}) + \frac{\Delta x}{f(\rho(x, t)) \sqrt{\mathbf{u}^\top M(x; \eta_0) \mathbf{u}}} \right]. \quad (10)$$

The optimal direction is

$$\mathbf{u}^*(x, t) = \arg \min_{\mathbf{u} \in U(x; s)} (\dots), \quad (11)$$

and the velocity is

$$\mathbf{v}(x, t) = f(\rho(x, t)) \mathbf{u}^*(x, t). \quad (12)$$

For the multi-stage multi-route model, each subpopulation (σ, r) uses its own target-specific potential $\phi_{\sigma, r}$ while sharing the same total-density congestion feedback. Thus, direction rules, geometric guidance, and density-dependent congestion are coupled in a single Bellman–conservation-law model.

D. Internal Entrance Capacity Control

Direction rules alone cannot represent the management action in which a channel is open but its entrance must be metered [59], [60], [62]. For each channel, we introduce maximum internal entrance rates

$$q = \{q_c^+(t), q_c^-(t)\}_{c=1}^C, \quad (13)$$

where $q_c^+(t)$ and $q_c^-(t)$ denote the maximum admitted rates at the reference forward and reverse entrances of channel c . These controls act on internal channel entrance fluxes rather than on external boundary inflows. In physical units, both the attempted flux $A_c^\pm(t)$ and the capacity bound $q_c^\pm(t)$ are entrance flow rates measured in ped/s.

The direction state and entrance capacity must satisfy the hard consistency constraints

$$\begin{aligned} s_c = +1 &\Rightarrow q_c^-(t) = 0, \\ s_c = -1 &\Rightarrow q_c^+(t) = 0, \\ s_c = \emptyset &\Rightarrow q_c^+(t) = q_c^-(t) = 0, \\ s_c = 0 &\Rightarrow q_c^+(t) + q_c^-(t) \leq \bar{q}_c. \end{aligned} \quad (14)$$

Let $A_c^\pm(t)$ be the attempted entrance flux induced by free movement. The actually admitted flux is

$$\hat{A}_c^\pm(t) = \min\{A_c^\pm(t), q_c^\pm(t)\}. \quad (15)$$

When $A_c^\pm(t) > 0$, the flux across the internal entrance is scaled by

$$\theta_c^\pm(t) = \frac{\hat{A}_c^\pm(t)}{A_c^\pm(t)}. \quad (16)$$

The excess demand $(A_c^\pm(t) - \hat{A}_c^\pm(t))_+$ remains upstream and is recorded as waiting or queueing mass. This treatment changes channel usage and risk distribution while preserving the mass balance of the conservation-law update.

E. Numerical Solver and Simulation Operator

The walkable region, obstacles, channel areas, and internal entrance interfaces are discretized on a uniform two-dimensional grid. At each time step, the total density is first computed from all stage-route subpopulations and used to update $f(\rho)$. For each subpopulation, the admissible direction set and mobility tensor are then constructed, and the corresponding discrete Bellman equation is solved. After the velocity direction is recovered, an upwind finite-volume update advances the conservation law, while internal entrance fluxes are modified by $\theta_c^\pm(t)$ [49]. Finally, stage transition terms are applied to update the subpopulation densities.

For any feasible control variable $z = (s, q)$, the complete numerical workflow defines a lower-level simulation operator

$$(\rho, \phi, \mathbf{v}, B, R) = \mathcal{S}(z; \hat{p}, \eta_0), \quad (17)$$

where B records entrance waiting or blocking and $R = (R_1, \dots, R_C)$ records realized cumulative throughputs of the channels. The channel balance metric below is computed from the realized throughputs R_c , not from the prescribed capacity limits $q_c^\pm(t)$.

F. Evaluation Metrics and Optimization Objective

1) *Efficiency, Safety, and Load Balance*: Following recent passenger-flow-control and crowd-evacuation simulation-optimization studies [38], [41], [62], the total travel time metric is

$$J_1(z) = \int_0^T \int_{\Omega_w} \rho(x, t) dx dt. \quad (18)$$

The high-density exposure metric is

$$J_2(z) = \int_0^T \int_{\Omega_w} \mathbf{1}_{\rho(x, t) > \rho_{\text{safe}}} dx dt. \quad (19)$$

The safety threshold is the density above which exposure is counted, and is fixed at $\rho_{\text{safe}} = 3.5$ ped/m² in both the four-channel and Bund scenarios. Let

$$R_c(T) = \int_0^T (\hat{A}_c^+(t) + \hat{A}_c^-(t)) dt \quad (20)$$

be the realized cumulative throughput of channel c . The channel load imbalance is

$$J_3(z) = \text{Var}(R_1(T), \dots, R_C(T)). \quad (21)$$

2) *Entrance Waiting and Control Smoothness*: Entrance waiting is penalized as

$$J_4(z) = \int_0^T \sum_{c=1}^C (B_c^+(t) + B_c^-(t)) dt. \quad (22)$$

For piecewise-constant capacity profiles, smoothness is measured by

$$J_5(z) = \sum_{c,k} \left[(q_{c,k+1}^+ - q_{c,k}^+)^2 + (q_{c,k+1}^- - q_{c,k}^-)^2 \right]. \quad (23)$$

Here k indexes the capacity-control time segment. J_4 discourages excessive entrance metering that creates upstream queues, while J_5 discourages abrupt high-frequency changes in capacity profiles [42], [59], [60], [62].

3) *Normalization and Scalar Objectives*: The raw metrics above have different physical units and numerical scales. Therefore, the reported scalar objectives are computed from dimensionless normalized quantities. Let

$$M_{\text{ref}} = M_0 + M_{\text{in}}, \quad T_e = T,$$

where M_0 is the initial crowd mass, M_{in} is the cumulative external inflow during $[0, T]$, and T_e is the evaluation horizon. Let $|\Omega_w|$ be the walkable area, $R_{\text{tot}} = \sum_{c=1}^C R_c(T)$, and

$$D_R = R_{\text{tot}}^2 \frac{C-1}{C^2}.$$

The basic normalized terms are

$$\begin{aligned} \bar{J}_1 &= \frac{J_1}{M_{\text{ref}} T_e}, & \bar{J}_2 &= \frac{J_2}{|\Omega_w| T_e}, \\ \bar{J}_3 &= \frac{J_3}{D_R + \varepsilon_R}, & \bar{J}_4 &= \frac{J_4}{M_{\text{ref}} T_e}. \end{aligned}$$

For segmented entrance-rate profiles, the smoothness term is normalized by the corresponding entrance-rate bounds:

$$\bar{J}_5 = \frac{1}{N_q} \sum_{c,k,\sigma \in \{+, -\}} \left(\frac{q_{c,k+1}^\sigma - q_{c,k}^\sigma}{\bar{q}_c^\sigma + \varepsilon_q} \right)^2.$$

Here N_q is the number of active finite differences, $\bar{q}_c^\sigma$ is the reference upper bound of the corresponding entrance, and ε_R and ε_q only avoid zero denominators. Thus, $\bar{J}_1$ is a relative system-residence measure. The term $\bar{J}_2$ is the average hard exposure or soft safety-risk intensity over the walkable space-time domain [7]. The term $\bar{J}_3$ is zero for perfectly balanced realized channel use and approaches one when all realized channel throughput concentrates on one channel.

The reported terms use scale factors

$$\tilde{J}_k = \frac{\bar{J}_k}{a_k}. \quad (24)$$

The main experiments use

$$(a_1, a_2, a_3, a_4, a_5) = (1, 10^{-3}, 1, 1, 1).$$

Therefore, the reported $\tilde{J}_2$ values are scaled normalized safety terms and should not be read directly as space-time percentages; the basic normalized value is $\bar{J}_2 = 10^{-3} \tilde{J}_2$. Given weights $\lambda = (\lambda_1, \dots, \lambda_5)$, the scalarized objective is

$$J(z) = \sum_{k=1}^5 \lambda_k \tilde{J}_k(z), \quad (25)$$

TABLE I
RAW UNITS OF THE OBJECTIVE TERMS AND THEIR NORMALIZERS

Term	Raw unit	Normalizer (same unit)	Reported
J_1	ped·s	$M_{\text{ref}}T_e$ (ped·s)	dimensionless
J_2	$\text{m}^2\cdot\text{s}$	$ \Omega_w T_e$ ($\text{m}^2\cdot\text{s}$)	dimensionless
J_3	ped^2	D_R (ped^2)	dimensionless
J_4	ped·s	$M_{\text{ref}}T_e$ (ped·s)	dimensionless
J_5	(ped/s) ²	$(\tilde{q}_c^c)^2$ (ped/s) ²	dimensionless

with the main-experiment setting

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = (1, 1, 1, 1, 0.1).$$

and the optimization problem is

$$\min_{z \in \mathcal{Z}} J(z), \quad z = (s, q). \quad (26)$$

The paper solves this fixed-weight scalarized management problem. Different weights can represent different management preferences, but the algorithm itself does not depend on one specific objective term being independently optimal. Although J , $J^{(3)}$, and $\tilde{J}_k$ are dimensionless ranking quantities, their interpretation is anchored by the normalization above [46], [63]. When physical interpretation is needed, the experiments also report raw or ratio-based diagnostics such as channel-flow shares, system mass, and density peaks. Table I lists the physical units of the raw objective terms and their normalizers; in each case the term and its normalizer share the same unit, so every reported $\tilde{J}_k$ is dimensionless.

For mechanism-verification experiments and fixed-strategy diagnostic experiments that only test direction rules, geometric guidance, or fixed entrance-rate responses, we use the reduced three-term objective

$$J^{(3)}(z) = \lambda_1 \tilde{J}_1(z) + \lambda_2 \tilde{J}_2(z) + \lambda_3 \tilde{J}_3(z). \quad (27)$$

In these experiments, the entrance-waiting term $\tilde{J}_4$ is either reported only as a diagnostic quantity or is not applicable, and the smoothness term $\tilde{J}_5$ is meaningful only for segmented capacity-profile optimization. Equivalently, reduced-objective experiments use $\lambda_4 = \lambda_5 = 0$. The numerical values of J and $J^{(3)}$ are not compared across experiment groups.

G. Hierarchical Constrained Mixed-Variable Black-Box Optimization

Since s is discrete, q is continuous or piecewise continuous, and each objective evaluation requires the expensive simulator $\mathcal{S}$, the resulting problem is an expensive mixed-variable black-box optimization problem [64], [65], [66]. We propose HCMBO, a Hierarchical Constrained Mixed-variable Black-box Optimization method, to exploit the structural relation between direction variables and capacity variables.

1) *Direction Candidates and Capacity Parameterization:* HCMBO first generates a direction candidate set $\mathcal{S}_{\text{cand}}$ satisfying operational and connectivity constraints. For a fixed direction configuration s , the entrance capacity profile is represented by a low-dimensional continuous vector $x \in [0, 1]^d$ and mapped to the actual capacity control by

$$q = T_s(x), \quad x \in [0, 1]^d. \quad (28)$$

The mapping T_s automatically applies direction masks, capacity upper bounds, closure constraints, and smoothness restrictions. For example, if a channel is set to reference forward one-way, the reverse entrance capacity is set to zero; if a channel is closed, both directional capacities are set to zero. Thus, the optimizer searches in the continuous x -space while all candidates are mapped to capacity profiles consistent with the discrete direction configuration.

2) *Structured Capacity Search:* For each direction candidate s , HCMBO performs structured search in the corresponding capacity parameter space. The search uses previously evaluated samples to construct surrogate information for objective values and feasibility, and preferentially selects capacity candidates that satisfy constraints and may improve the incumbent objective. This hierarchical design avoids blind sampling in the full mixed-variable space and separates discrete direction structure from continuous capacity structure.

Multi-fidelity evaluation can be used during search. Low- or medium-fidelity simulations help identify clearly poor capacity profiles, whereas high-fidelity simulations are reserved for final candidate rechecking and ranking. To avoid discarding good long-horizon strategies due to short-horizon bias, low-fidelity results are used only as auxiliary evaluation signals and do not directly determine the final optimum.

3) *High-Fidelity Rechecking and Final Selection:* All finalist candidates are re-evaluated under a unified high-fidelity (HF) setting. Let $\mathcal{C}_{\text{HF}}$ be the candidate set selected for high-fidelity rechecking. The final output is

$$z^* = \arg \min_{z \in \mathcal{C}_{\text{HF}}} J_{\text{HF}}(z). \quad (29)$$

This mechanism prevents surrogate error, low-fidelity bias, or intermediate search ranking from directly becoming the final performance claim.

Algorithm 1 gives the reproducible HCMBO procedure used in the experiments. The loop terminates after the finite optimization budget has been allocated across direction candidates, and the reported optimum is selected only after the high-fidelity rechecking stage.

Compared with generic mixed-variable optimization, HCMBO uses the dependence between direction choices and feasible capacity profiles. Once s is fixed, the mapping T_s removes capacity variables that conflict with the selected direction rules and bounds the remaining entrance-rate profiles. This reduces invalid candidates and allocates the simulation budget to policies that satisfy the operational constraints.

IV. NUMERICAL EXPERIMENTS

We validate the proposed model and optimizer on a four-channel scenic-platform scenario abstracted from the Bund sightseeing platform setting. The scenario consists of an inflow region, a Bund sightseeing platform and exit region, a central wall, and four connecting channels: upper, middle, lower-middle, and bottom channels. Throughout this paper, channel denotes a model flow entity indexed by c , whereas passage refers to a real-world walkway at the Bund site that a channel abstracts. The experiments are organized as mechanism validation, directional-response analysis, entrance-rate response, horizontal optimizer comparison, and HCMBO ablation.

Algorithm 1 HCMBO for Joint Direction and Entrance-Rate Control

Require: Direction constraints Ω_s ; simulator $\mathcal{S}$; objective J ; gate-rate bounds $\bar{q}$; budget B ; initial design size n_0 ; pool size P ; lower confidence bound (LCB) coefficient β ; high-fidelity shortlist size K .

Ensure: Best high-fidelity policy z^* .

- 1: Generate $\mathcal{S}_{\text{cand}}$ by enumerating $s \in \Omega_s$ and retaining only configurations satisfying open-channel and connectivity constraints.
 - 2: Rank $\mathcal{S}_{\text{cand}}$ using the prescribed priority templates and direction proxy score, and then truncate it to the direction-candidate limit.
 - 3: Set $b_s = \lfloor B/|\mathcal{S}_{\text{cand}}| \rfloor$ and assign one extra evaluation to the first $B \bmod |\mathcal{S}_{\text{cand}}|$ directions.
 - 4: Initialize the global evaluation log $\mathcal{D} \leftarrow \emptyset$.
 - 5: **for all** $s \in \mathcal{S}_{\text{cand}}$ **do**
 - 6: Construct $T_s : [0, 1]^{d_s} \rightarrow q$ by applying the direction mask, gate bounds $\bar{q}$, and time segmentation.
 - 7: Initialize $\mathcal{D}_s \leftarrow \emptyset$ and generate X_s^0 from high-, medium-, and low-rate profiles plus random fill-up points.
 - 8: **for all** $x \in X_s^0$ **do**
 - 9: **if** $|\mathcal{D}_s| < b_s$ **then**
 - 10: Evaluate $z = (s, T_s(x))$ with the optimization-fidelity simulator and append $(x, z, J(z))$ and feasibility metrics to $\mathcal{D}_s$ and $\mathcal{D}$.
 - 11: **end if**
 - 12: **end for**
 - 13: **while** $|\mathcal{D}_s| < b_s$ **do**
 - 14: Let x_s^{best} be the evaluated point with the lowest objective in $\mathcal{D}_s$.
 - 15: Draw $\mathcal{P}_s$ from P uniform samples in $[0, 1]^{d_s}$ and $\max(4, \lfloor P/4 \rfloor)$ Gaussian perturbations of x_s^{best} , clipped to $[0, 1]^{d_s}$.
 - 16: Set the kernel length scale $\ell_s = \max\{0.18, 1/\sqrt{d_s}\}$.
 - 17: **for all** $u \in \mathcal{P}_s$ **do**
 - 18: Estimate $\hat{\mu}_s(u)$ by weighted averaging with $w_i(u) = \exp(-\|x_i - u\|_2^2 / (2\ell_s^2))$ over $(x_i, z_i, J_i) \in \mathcal{D}_s$.
 - 19: Set $\hat{\sigma}_s(u) = \min\{1, \min_{(x_i, z_i, J_i) \in \mathcal{D}_s} \|u - x_i\|_2 / \sqrt{d_s}\}$.
 - 20: Compute $a_s(u) = \hat{\mu}_s(u) - \beta \text{std}(\mathcal{D}_s) \hat{\sigma}_s(u)$.
 - 21: **end for**
 - 22: Select $x_{\text{next}} \leftarrow \arg \min_{u \in \mathcal{P}_s} a_s(u)$.
 - 23: Evaluate $z_{\text{next}} = (s, T_s(x_{\text{next}}))$ and append the result to $\mathcal{D}_s$ and $\mathcal{D}$.
 - 24: **end while**
 - 25: **end for**
 - 26: Select the top K unique controls $\mathcal{C}_{\text{HF}}$ from $\mathcal{D}$ by the optimization-fidelity objective.
 - 27: **for all** $z \in \mathcal{C}_{\text{HF}}$ **do**
 - 28: Re-evaluate z under the unified high-fidelity simulator setting to obtain $J_{\text{HF}}(z)$.
 - 29: **end for**
 - 30: **return** $z^* = \arg \min_{z \in \mathcal{C}_{\text{HF}}} J_{\text{HF}}(z)$.
-

A. Experimental Setup

The domain is discretized on a 128×96 grid. To report results in physical units, simulation quantities are mapped through a representative calibration based on three reference scales: a grid cell size of $\Delta x = 1.0$ m, so that the domain spans $128 \text{ m} \times 96 \text{ m}$; a free-flow walking speed of 1.5 m/s ; and a jam density of 5.0 ped/m^2 . The induced time step is then $\Delta t = 0.2 \text{ s}$, and pedestrian counts follow by integrating density over physical area, so that a fully occupied cell holds

$\rho_{\text{max}} \Delta x^2 = 5 \text{ ped}$. As a coherence check, a fully packed domain of this size would hold at most $\rho_{\text{max}} \times 128 \times 96 \approx 6.1 \times 10^4 \text{ ped}$, which bounds all reported pedestrian counts. These physical units are adopted only for readability and reflect a representative assumption; they do not represent any specific venue.

The channel width is 4 m , and the controlled internal entrance interface width is 16 m . The free-flow speed is $v_{\text{max}} = 1.5 \text{ m/s}$, the maximum density is $\rho_{\text{max}} = 5.0 \text{ ped/m}^2$, the initial density is $\rho_{\text{init}} = 2.2 \text{ ped/m}^2$, and the Courant–Friedrichs–Lewy (CFL) factor is 0.9 .

Mechanism and response experiments use the corresponding validation horizons. The optimizer comparison uses high-fidelity rechecks with 1600 simulation steps, a horizon of 320 s (i.e. $\Delta t = 0.2 \text{ s}$), and Bellman updates every five steps. Final optimizer rankings are based on the high-fidelity objective in Section III-F. Unless otherwise stated, optimization experiments are ranked by the full five-term scalar objective J after high-fidelity rechecking.

Mechanism-validation, directional-response, fixed-rate response, and supplementary transfer diagnostics use the reduced objective $J^{(3)} = \lambda_1 \tilde{J}_1 + \lambda_2 \tilde{J}_2 + \lambda_3 \tilde{J}_3$. For these reduced-objective experiments, $\tilde{J}_4$ is only an entrance-waiting diagnostic or is not applicable, and $\tilde{J}_5$ is not applicable. Equivalently, $\lambda_4 = \lambda_5 = 0$. Therefore, the numerical values of J and $J^{(3)}$ should not be compared across experiment groups. Comparisons are meaningful only within the same objective definition, normalization rule, and weight setting. A high-fidelity best candidate is called feasible if the density-cap removal mass is no more than 2% of the reference mass.

Unless otherwise stated, cumulative flow, waiting mass, unreleased attempted flow, and system mass are reported in pedestrians (ped), obtained by integrating density over the physical cell area. Mass-time quantities are reported in ped·s. Counterflow area-time is reported in $\text{m}^2 \cdot \text{s}$. Channel-flow shares are computed from realized cumulative channel throughputs, $R_c(T) / \sum_j R_j(T)$. Scalar objectives and normalized terms are dimensionless. Raw cumulative quantities are rounded to three significant digits in the text, while tables retain four decimals for dimensionless objective terms.

B. Mechanism Validation

We first verify whether the two local modeling components have the intended behavioral meanings. The fixed anisotropic mobility tensor $M(x; \eta_0)$ should reshape the channel attraction domain and induce entrance pre-alignment, whereas the admissible direction set $U(x; s)$ should remove prohibited directions and reroute flow through feasible channels. When M is applied to the middle channel, its flow share increases from 48.78% in the baseline to 100.00%. When a one-way rule excludes the originally dominant middle-channel direction, the middle-channel share becomes zero and the flow is redistributed to the upper, lower-middle, and bottom channels with shares 19.25%, 68.76%, and 12.00%, respectively. For the bidirectional validation case, we use two diagnostic quantities. The reverse-direction mass time (RMT) is

$$\text{RMT} = \sum_n \sum_{i \in \Omega_c} \rho_i^n \mathbf{1}\{\mathbf{u}_i^n \cdot \boldsymbol{\tau}_c < 0\} \Delta x^2 \Delta t,$$

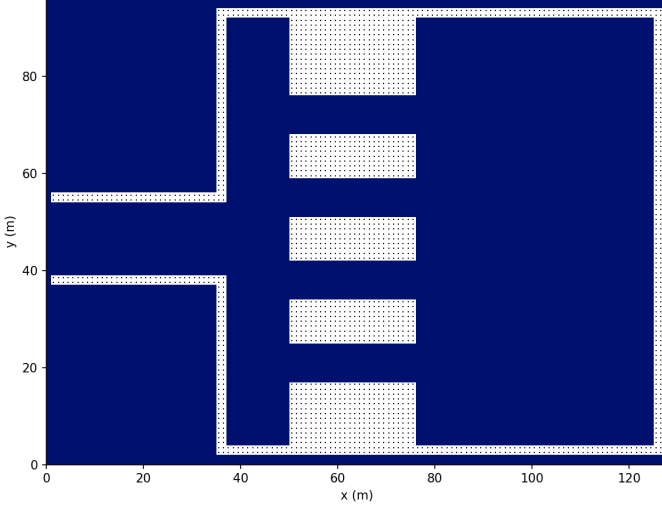


Fig. 3. Four-channel scenic-platform simulation scenario used in the numerical experiments.

which measures the cumulative reverse-direction mass-time (in ped·s) moving against the prescribed channel direction. The counterflow area-time (CAT) is

$$\text{CAT} = \sum_n \sum_{i \in \Omega_c} \mathbf{1}\{\rho_{i,+}^n > 0, \rho_{i,-}^n > 0\} \Delta x^2 \Delta t,$$

which measures the cumulative area-time (in $\text{m}^2\cdot\text{s}$) during which opposite-direction subpopulations coexist in the same channel region. In the baseline bidirectional inflow test, the westbound middle-channel share, RMT, and CAT are 73.18%, 3.33×10^3 ped·s, and 2.43×10^3 $\text{m}^2\cdot\text{s}$, respectively. After imposing the eastbound one-way rule, all three quantities are reduced to zero.

Fig. 4 compares guidance applied to different channels. In each case, the dominant flow channel and local density hotspot move with the guided channel. This supports using the Bellman-conservation-law simulator as the lower-level evaluator for subsequent strategy optimization.

C. Directional-Control Response

We next examine whether the discrete direction configuration s creates nontrivial system-level trade-offs. The response experiment uses thirteen interpretable direction settings: an all-free baseline, four single-entry cases, four single-return cases, and four one-channel-closure cases. The direction code is ordered as $[T, M, LM, B]$, representing the top, middle, lower-middle, and bottom channels, where E, W, F , and X denote eastbound entry, westbound return, bidirectional free channel, and closure. Policy labels use baseline (BSL), single-entry (SE), and single-return (SR), with suffixes identifying the affected channel.

Fig. 5 and Table II show that no single direction configuration dominates all objectives. This directional-response experiment summarizes the three reported terms by $J^{(3)}$; since no entrance-capacity curve is optimized, $\tilde{J}_4$ and $\tilde{J}_5$ are not used for ranking. C8 (SR-LM, EEWE) obtains the lowest reduced objective $J^{(3)} = 0.5102$, which is only 0.84% lower than the all-free baseline C1. C1 gives the lowest safety exposure

TABLE II
NON-DOMINATED DIRECTIONAL-CONTROL CASES

Case	Policy	Code	$\tilde{J}_1$	$\tilde{J}_2$	$\tilde{J}_3$	$J^{(3)}$	Main characteristic
C1	BSL	FFFF	0.3535	0.00785	0.1531	0.5145	Lowest exposure
C3	SE-M	WEWW	0.3141	0.01680	0.2741	0.6049	Best efficiency but risk-concentrated
C4	SE-LM	WWEW	0.3335	0.01662	0.3131	0.6632	Efficient but load-imbalanced
C6	SR-T	WEEE	0.4029	0.01174	0.1335	0.5482	High return throughput
C8	SR-LM	EEWE	0.3395	0.01311	0.1576	0.5102	Lowest reduced objective
C9	SR-B	EEEW	0.4377	0.01344	0.1138	0.5649	Most balanced channel loads

All objective terms in this table are dimensionless scaled normalized quantities. The reduced objective uses $\lambda_4 = \lambda_5 = 0$ and is used only for directional-response diagnosis; it is not directly comparable with the full five-term objective in Table IV.

$\tilde{J}_2 = 0.00785$, C3 gives the best efficiency term $\tilde{J}_1 = 0.3141$, and C9 gives the best load-balance term $\tilde{J}_3 = 0.1138$. This non-monotonic response makes manual rule selection from a small set insufficient and motivates automatic mixed-variable optimization.

D. Entrance-Rate Response

Direction rules alone cannot represent the case in which a channel remains open but its internal entrance must be metered. We therefore fix $[T, M, LM, B] = [F, E, W, F]$ and scan four internal entrance-rate levels: no limit, high, medium, and low. The rate upper bound acts on internal entrance crossing flow. When attempted flow exceeds the upper bound, only the admitted flow crosses the interface, and the unreleased attempted flow remains upstream as waiting mass. The entrance binding-time ratio is defined as the fraction of time steps in which attempted entrance flow exceeds the rate upper bound and creates unreleased attempted flow. The unreleased attempted flow is

$$U_{\text{rel}} = \int_0^T \sum_{c=1}^C \left[(A_c^+(t) - \hat{A}_c^+(t))_+ + (A_c^-(t) - \hat{A}_c^-(t))_+ \right] dt,$$

in pedestrian-equivalent demand (ped·eq.); because held-back mass re-attempts the interface at later steps, U_{rel} is a cumulative time integral of repeatedly denied demand rather than an instantaneous queue length, and may exceed the number of distinct waiting pedestrians. Its demand ratio is $r_{\text{rel}} = U_{\text{rel}} / \int_0^T \sum_c (A_c^+(t) + A_c^-(t)) dt$. This scan is a fixed-rate diagnostic experiment rather than a final optimizer-ranking experiment; the table therefore reports $J^{(3)}$, treats entrance waiting as a diagnostic quantity, and sets the smoothness contribution to zero because each policy uses a fixed rate level.

Table III and Fig. 6 show that tightening the rate upper bound does not simply improve safety. From the no-limit case to the low level, unreleased attempted flow increases from 0 to 1.71×10^3 ped·eq., corresponding to $r_{\text{rel}} = 92.1\%$ of attempted entrance demand. The binding ratio increases from

U+M Configuration Sensitivity: Guided Channel Comparison

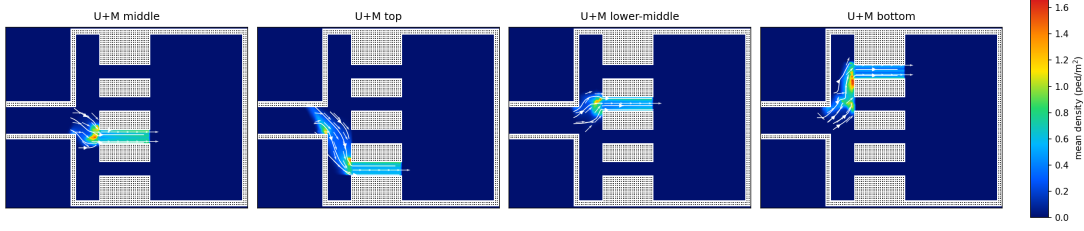


Fig. 4. Spatial transferability of the combined $U+M$ mechanism. Moving the controlled channel induces corresponding migration of dominant flow channels, streamlines, and density hotspots.

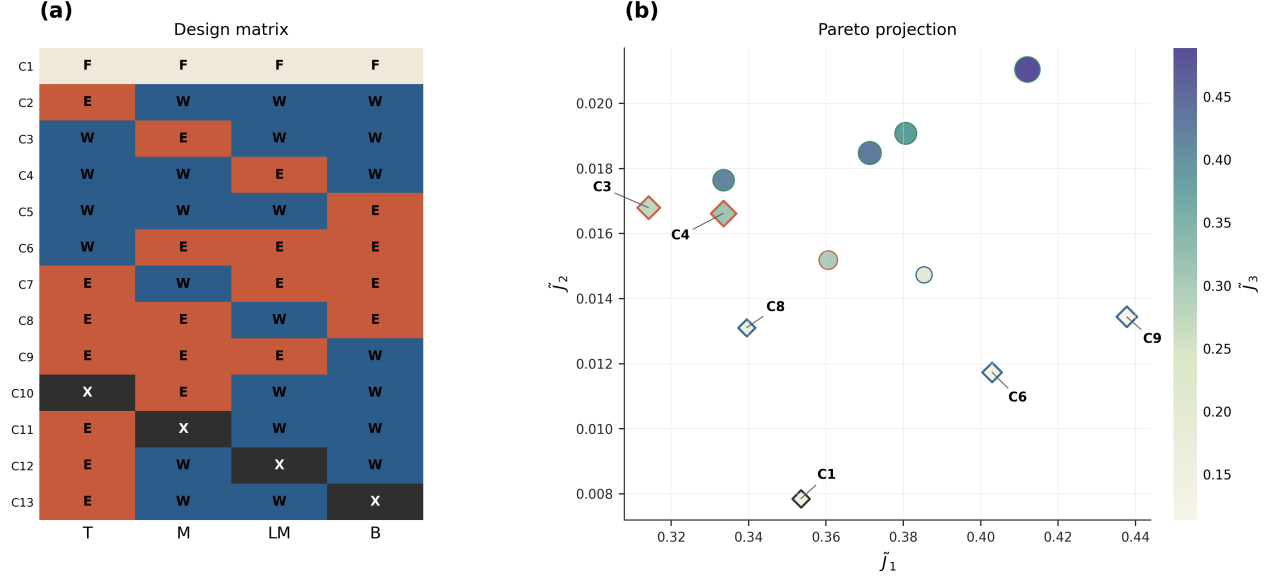


Fig. 5. Directional-control design matrix and objective-space trade-off. Color encodes $\tilde{J}_3$, marker size encodes the reduced dimensionless diagnostic objective $J^{(3)}$, and diamonds indicate non-dominated cases.

TABLE III
RESPONSE UNDER INTERNAL ENTRANCE-RATE
UPPER-BOUND LEVELS

Level	$\tilde{J}_1$	$\tilde{J}_2$	$\tilde{J}_3$	$J^{(3)}$	U_{rel}	r_{rel}	Binding ratio
No limit	0.6274	0.174	0.8838	1.6851	0.00	0	0.000
High	0.6262	0.358	0.8814	1.8652	965.60	70.8%	0.605
Medium	0.6228	0.443	0.8620	1.9276	1317.04	82.6%	0.666
Low	0.6181	0.466	0.8493	1.9337	1711.56	92.1%	0.721

U_{rel} is reported in pedestrian-equivalent demand (ped-eq.), a cumulative time integral of denied entrance demand and not an instantaneous queue length. The safety column is a scaled normalized safety term, not a direct space-time percentage.

0 to 0.721, and the scaled normalized safety term $\tilde{J}_2$ increases from 0.174 to 0.466. Thus, q is a genuine continuous control variable that changes system response and must be optimized jointly with s .

E. Horizontal Optimization Comparison

The horizontal comparison evaluates Prior Baseline, Random Search, Pure Simulated Annealing (Pure SA), Enumeration plus Differential Evolution (Enum-DE), TPE-Mixed

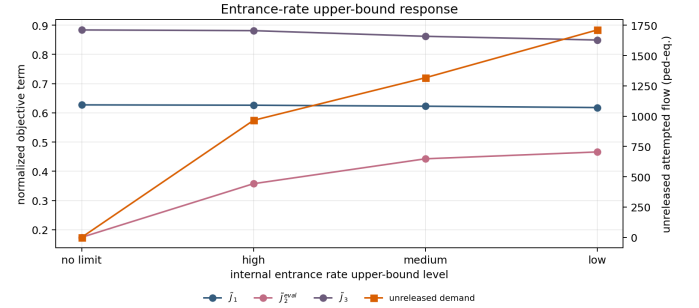


Fig. 6. Objective-term response under internal entrance-rate upper-bound levels. Tightening the upper bound changes safety exposure, load balance, and unreleased attempted flow; the latter is reported in pedestrian-equivalent demand (ped-eq.).

BO, and HCMBO. All optimizers use $B = 400$ candidate evaluations and top-10 high-fidelity rechecks over seeds 11, 23, 37, 51, and 73.

Table IV reports the smallest mean, median, and best objective for HCMBO, together with the highest feasible rate among the tested methods. Its mean objective is 3.9% lower than TPE-Mixed BO, since $(2.7403 - 2.6338) / 2.7403 = 3.9\%$.

TABLE IV
HIGH-FIDELITY PERFORMANCE UNDER THE FULL FIVE-TERM
DIMENSIONLESS OBJECTIVE

Method	Mean objective	Std.	Median	Best	Worst	Feasible rate
HCMBO	2.6338	0.3106	2.5065	2.4495	3.1851	100%
TPE-Mixed BO	2.7403	0.2475	2.6288	2.5180	3.1345	80%
Pure SA	2.9248	0.1386	2.9203	2.7468	3.1049	60%
Random Search	3.0112	0.1319	2.9797	2.8821	3.2287	40%
Enum-DE	3.0597	0.1932	3.1004	2.7329	3.2500	40%
Prior Baseline	5.4774	0.0000	5.4774	5.4774	5.4774	0%

The reported objective is the full dimensionless scalar objective $J = \sum_{k=1}^5 \lambda_k \tilde{J}_k$ after unified high-fidelity rechecking. These values are not directly comparable with the reduced diagnostic objective $J^{(3)}$ in Tables II, III, and VII.

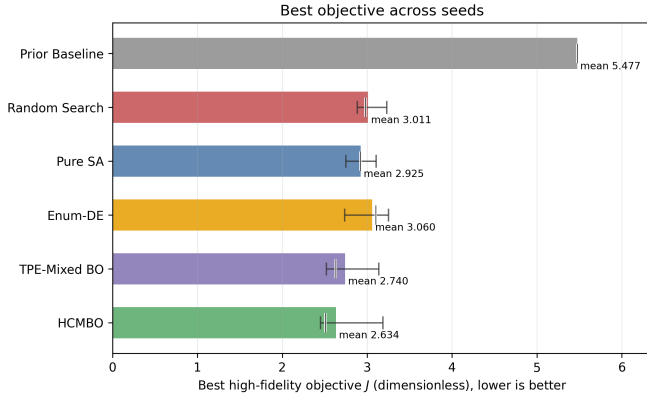


Fig. 7. Best high-fidelity full objective J across seeds. The objective is dimensionless and lower is better. Range bars show the seed-wise spread, and the short vertical mark indicates the median.

Because this is a normalized scalar objective, the percentage reduction reflects the fixed-weight management score, not a direct reduction in travel time or density exposure alone. In paired comparisons, HCMBO has a lower objective than TPE-Mixed BO in four of five seeds, with an average objective difference of -0.1066 . It is also lower than Random Search in all five seeds and lower than Pure SA and Enum-DE in four of five seeds. The remaining high HCMBO objective is mainly caused by a larger safety-exposure term, so later versions should add more explicit safety constraints during search.

Figs. 7–9 compare the seed-wise objectives, convergence traces, and selected entrance-rate profiles. TPE-Mixed BO decreases quickly in the early evaluations, whereas HCMBO continues to reduce the best-so-far objective during the middle and late stages. The control profiles indicate that methods with the same direction code can still choose different segment-wise rate allocations, leading to different final objectives.

F. HCMBO Ablation

The ablation experiments test the contribution of each control and search component. Optimizing only directions with a fixed high rate gives a best high-fidelity objective of 3.3751, and optimizing only rates under prior directions gives 3.3140;

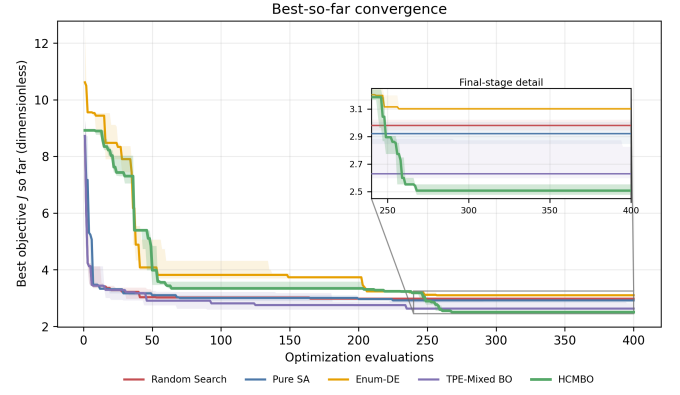


Fig. 8. Best-so-far convergence curves under the same 400-evaluation budget. The inset highlights the final-stage separation among methods.

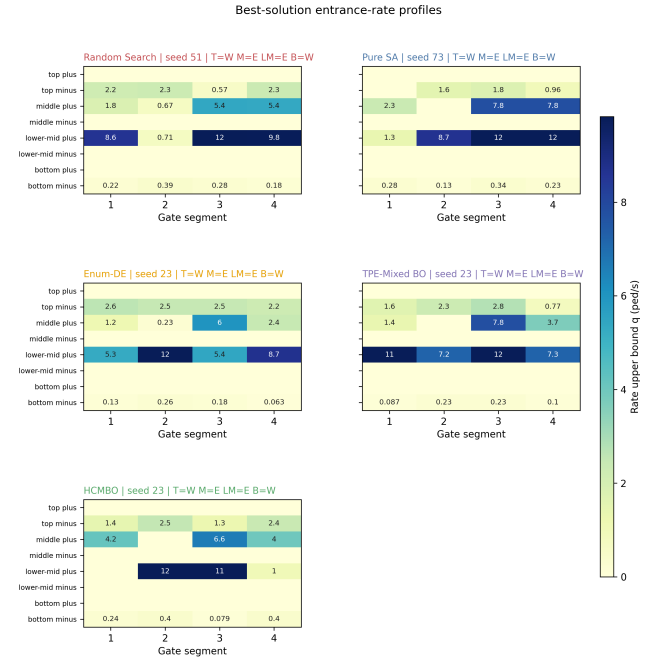


Fig. 9. Best-solution entrance-rate profiles for Random Search, Pure SA, Enum-DE, TPE-Mixed BO, and HCMBO. The common direction structure is $[T, M, LM, B] = [W, E, E, W]$ for several methods, but the segment-wise rate allocations differ substantially.

both are higher than the joint optimization result. Short-horizon low-fidelity hard screening raises the best objective to 2.9441 and selects an infeasible best candidate, because early simulations do not reliably capture long-horizon entrance waiting and load redistribution. Removing the entrance-waiting penalty slightly lowers the scalar objective to 2.6015, but increases unreleased attempted flow to approximately 1.30×10^4 ped-eq. The waiting term should therefore remain in the management objective. In the search-mechanism variants, lower confidence bound (LCB) denotes the acquisition rule, and random-forest-style (RF-style) denotes the constrained Bayesian-optimization surrogate variant.

Table V shows that the final HCMBO mainline has the lowest scalar objective among the tested variants. Queue-

TABLE V
SEARCH-MECHANISM ABLATION UNDER SEED 23

Variant	Best objective	Feasible rate	$\tilde{J}_2$	Unreleased flow (ped-eq.)	Main meaning
HCMBO	2.4495	100%	1.6759	1.02×10^4	Structured direction-wise rate search
Queue-aware LCB	2.6162	100%	1.8521	9.64×10^3	Queue-aware acquisition
RF-style constrained BO	2.6162	100%	1.8521	9.64×10^3	Constraint surrogate variant
Adaptive racing	2.6561	100%	1.9474	1.28×10^4	Stage-wise budget concentration
Adaptive racing + queue-aware	2.6683	100%	1.7862	1.06×10^4	Combined racing and queue signal
Trust-region search	3.0317	100%	1.9224	1.59×10^4	Local trust-region search

Unreleased flow is the cumulative unreleased attempted entrance flow over the full horizon, in pedestrian-equivalent demand (ped-eq.); it integrates repeatedly denied demand and is not an instantaneous queue length.

aware LCB and RF-style constrained BO reduce unreleased attempted flow to approximately 9.64×10^3 ped-eq., but their scalar objective increases to 2.6162. Thus, queue information is useful for executable and safety-aware extensions, whereas the final mainline should retain structured direction-wise rate search as the primary budget allocation mechanism.

G. Transfer Validation on a Bund-Inspired Scenario

The preceding experiments use a standardized four-channel platform scenario to verify modeling mechanisms and optimization behavior under controlled conditions. As a supplementary validation, we further construct a simplified abstract scenario inspired by the spatial form of the Shanghai Bund waterfront. This experiment does not claim to reproduce all architectural details of the real site. Instead, it preserves the Bund sightseeing platform, the lateral passage connections, the major inflow location, and the dominant movement direction, so as to examine whether an HCMBO-derived policy can still change the load distribution in a more realistic spatial outline.

The computational grid has resolution 512×224 . Because this scenario preserves the real Bund outline, its physical scale is calibrated from georeferenced control points that match the scene coordinates to satellite latitude and longitude; a four-point affine fit gives 0.435 m per scene unit with sub-1.5 m residual, which corresponds to a physical cell size of about 1.75 m. The 512×224 grid therefore spans approximately $894 \text{ m} \times 391 \text{ m}$, the induced time step is $\Delta t \approx 0.44 \text{ s}$, and pedestrian counts again follow by integrating density over physical area (a cell area of about 3.0 m^2). Pedestrians enter from the outer-side entrance, pass through the middle connecting channels, and move toward the Bund sightseeing platform and its central target region. The inflow-rate multiplier is 3.0, corresponding to an actual inflow rate of about 3.35 ped/s; the maximum density is $\rho_{\max} = 4.0 \text{ ped/m}^2$, and the stage-switching rate is $\kappa_{\sigma,r} = 32$ in simulation time units. The experimental channels `channel_1`, `channel_2`,

TABLE VI
EFFECTIVE ENTRANCE-RATE UPPER BOUNDS IN THE BUND TRANSFER EXPERIMENT

Experimental channel	Local passage name	Entrance side	Segment 1 (ped/s)	Segment 2 (ped/s)
<code>channel_1</code>	Bund Passage Forward 3	entrance	3.45	8.38
<code>channel_1</code>	Bund Passage Reverse 3	entrance	3.17	0.00
<code>channel_2</code>	Bund Passage Forward 4	entrance	8.38	2.36
<code>channel_2</code>	Bund Passage Reverse 4	entrance	0.00	0.00
<code>channel_3</code>	Bund Passage Forward 5	entrance	0.00	0.00
<code>channel_3</code>	Bund Passage Reverse 5	entrance	3.57	6.95
<code>channel_4</code>	Bund Passage Forward 6	entrance	5.28	3.03
<code>channel_4</code>	Bund Passage Reverse 6	entrance	3.09	2.96

`channel_3`, and `channel_4` correspond to the locally named Bund Passages 3, 4, 5, and 6, respectively. Both the controlled and uncontrolled groups use 1600 steps over a physical horizon of about 698 s. For the controlled group, the transferred entrance-rate bounds are multiplied by a capacity-scale factor of 3, providing a less restrictive entrance-capacity setting.

The controlled group uses the best HCMBO policy obtained from the multi-direction high-fidelity recheck. Its reduced high-fidelity objective in the optimization stage is 0.7350, and its direction structure leaves the upper and lower channels bidirectional while assigning opposite dominant directions to the two middle channels. Table VI reports the effective two-segment entrance-rate upper bounds after applying the capacity-scale factor of 3. The rate upper bounds are measured in ped/s.

Table VII compares the controlled and uncontrolled groups under the same high-fidelity protocol. It separates dimensionless normalized terms from raw simulation diagnostics. The reduced objective $J^{(3)}$ decreases from 1.0140 to 0.7410, corresponding to an absolute reduction of approximately 0.2730 and a relative reduction of 26.9%. The largest reduction comes from the load-balance term $\tilde{J}_3$, which decreases by 51.9%, from 0.6043 to 0.2908. In contrast, $\tilde{J}_1$ increases by 8.7% and $\tilde{J}_2$ increases by 25.7%. This indicates that the transferred policy changes how pedestrians use the channels, rather than merely reducing all density values uniformly.

The channel-flow shares in Table VIII further explain this reduction. In the uncontrolled group, 82.30% of the channel flow concentrates on `channel_4`, i.e., Bund Passage 6. After applying the HCMBO policy, the flow is redistributed mainly to `channel_2` and `channel_3`, whose combined share reaches approximately 91.34%.

Fig. 11 visualizes the density evolution at approximately 22, 153, 393, and 654 s. Both groups initially form a local density concentration near the right-side target region of the Bund sightseeing platform, reflecting the same inflow and target-attraction structure. At later stages, the uncontrolled case remains more concentrated around the terminal channel



Fig. 10. Real Bund scene and its simplified simulation abstraction. The abstraction retains the Bund sightseeing platform boundary, the main entrance, and the passage connections used for control validation. The satellite basemap was obtained from Google Maps. The right-side local photographs were shot in the Shanghai Bund waterfront passage area by the Tongji Public Safety Experimental Team, June 21, 2026.

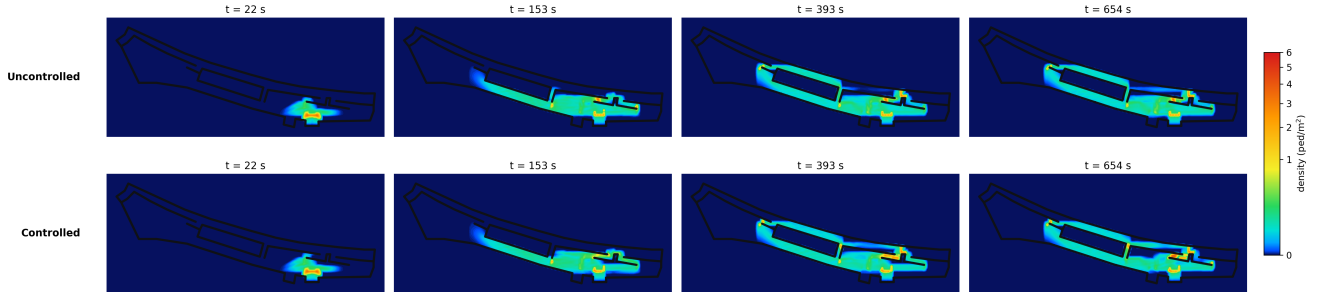


Fig. 11. Density evolution comparison between uncontrolled and controlled cases at approximately 22, 153, 393, and 654 s. The controlled policy redistributes the later-stage flow from the terminal channel toward the middle-channel group.

and the adjacent Bund sightseeing platform area, whereas the controlled case develops a more stable flow band through the middle-channel group. This visual evidence is consistent with the channel-share statistics and confirms that the main effect of the HCMBO policy is load redistribution across channels.

The transfer experiment exposes a trade-off. Compared with the uncontrolled group, the controlled group has a higher efficiency term ($\bar{J}_1$ increases from 0.3813 to 0.4145) and a slightly higher safety-exposure term ($\bar{J}_2$ increases from 0.0284 to 0.0357). Its cumulative sink flow is also smaller (1.33×10^3 vs. 1.67×10^3 ped), and its final system mass is larger (1.53×10^3 vs. 1.20×10^3 ped). The transferred HCMBO policy therefore should not be read as a uniform safety or efficiency improvement. Under the present reduced-objective weights, it lowers $J^{(3)}$ mainly by redistributing channel load,

while giving up part of exit throughput and travel efficiency. The capacity-scale factor of 3 reduces unreleased attempted flow to 364 ped-eq., but it does not remove the efficiency cost. A deployment-oriented version should add explicit throughput constraints and waiting upper bounds, and should test several objective-weight settings before selecting a policy.

The real Bund waterfront operates at crowd scales far beyond the demonstration level of this section. According to official holiday reports, the instantaneous on-site crowd in the Bund waterfront area peaked at 5.7×10^4 pedestrians during the 2025 National Day holiday [67] and reached 7.1×10^4 during the 2019 holiday [20]. The present experiment deliberately adopts a lower demand setting so that the transfer of control mechanisms and optimized policies can be examined under controlled conditions, rather than reproducing peak holiday

TABLE VII
CONTROLLED AND UNCONTROLLED RESULTS IN THE BUND TRANSFER EXPERIMENT

Dimensionless term	Controlled	Uncontrolled	Relative change
$J^{(3)}$	0.7410	1.0140	-26.9%
$\tilde{J}_1$	0.4145	0.3813	+8.7%
$\tilde{J}_2$	0.0357	0.0284	+25.7%
$\tilde{J}_3$	0.2908	0.6043	-51.9%
Raw diagnostic	Controlled	Uncontrolled	Unit
Final cumulative sink flow	1.33×10^3	1.67×10^3	ped
Final cumulative inflow	2.32×10^3	2.34×10^3	ped
Final system mass	1.53×10^3	1.20×10^3	ped
Peak density maximum	≈ 4.0	≈ 4.0	ped/m ²
Domain-mean density (incl. empty space)	3.47×10^{-3}	3.21×10^{-3}	ped/m ²
Attempted entrance flow	1.49×10^3	—	ped-eq.
Actual entrance flow	1.13×10^3	—	ped
Unreleased attempted flow	364	—	ped-eq.

Sink flow, inflow, system mass, and actual entrance flow are pedestrian counts (ped). Attempted and unreleased entrance flow are cumulative pedestrian-equivalent demand (ped-eq.): they integrate repeatedly denied demand over the horizon and may exceed the number of distinct waiting pedestrians. Entries marked — are not applicable because the uncontrolled case activates no internal metering interface. The domain-mean density is averaged over the full walkable domain, including empty space, and is therefore much smaller than the local peak.

TABLE VIII
CHANNEL-FLOW SHARES IN THE BUND TRANSFER EXPERIMENT

Channel	Local name	Controlled	Uncontrolled
channel_1	Bund Passage 3	2.83%	0.32%
channel_2	Bund Passage 4	30.38%	1.35%
channel_3	Bund Passage 5	60.96%	16.03%
channel_4	Bund Passage 6	5.83%	82.30%

Channel-flow shares are computed as $R_c(T) / \sum_{j=1}^C R_j(T)$, where $R_c(T)$ is the realized cumulative admitted throughput of channel c during the simulation horizon.

loads.

We emphasize that the computational cost of the macroscopic model is determined solely by the grid resolution and the number of simulation steps, and is independent of the number of pedestrians present: crowd size enters the model only through the magnitude of the density field and the entrance inflow rates. Scaling from the demonstration level to a 50,000-pedestrian operating condition therefore requires no change to the model structure, the numerical scheme, or the grid; only demand-side parameters change. Table IX contrasts the two operating conditions. The spatial extent, grid resolution, time step, and per-step cost are identical in both conditions; only the instantaneous population, the entrance inflow rates, and the horizon differ, and among these only the horizon increases the total cost, linearly in the number of steps.

From a dynamics viewpoint, a crowd of 5×10^4 – 7×10^4 pedestrians corresponds to an average walkable density of roughly 0.3–0.6 ped/m², far below the jam density $\rho_{\max} = 4.0$ ped/m², whereas the local peak density in this section already reaches $\rho_{\max}$; that is, the saturated congested branch of the fundamental diagram is exercised by the present experiments. Scaling up therefore enlarges the spatial extent of high-

density regions rather than pushing the model into a density regime untouched by the experiments. The proposed framework thus supports control analysis at the 50,000-pedestrian scale in terms of both computational feasibility and dynamical coverage; a full-region peak validation with field-calibrated demand profiles is left for future work.

V. CONCLUSION

This paper presents a unified macroscopic crowd management framework tailored for high-density metropolitan tourism hotspots, which seamlessly integrates passage-direction regulation, anisotropic geometric guidance, multi-stage route choice behavior, and internal entrance-rate control into a single Bellman-conservation-law operator. By tightly coupling a fully interpretable first-principles crowd flow simulator with the proposed Hybrid Constrained Multi-output Bayesian Optimization (HCMBO) optimizer, the proposed framework transforms the traditionally experience-driven, ad-hoc crowd control practice into a quantitative, fully reproducible, and operational-budget-aware decision-making pipeline.

Three core findings stand out from the extensive numerical experiments. First, all embedded modeling components behave exactly as designed, and the two proposed control dimensions are fundamentally complementary rather than interchangeable: direction settings induce genuinely non-monotonic trade-offs among traffic efficiency, safety exposure, and network load balance, while entrance-rate limits reshape the distribution of unreleased attempted flow, binding-time ratios, and local spatiotemporal risk patterns. Second, the proposed HCMBO algorithm consistently delivers the strongest overall performance across all tested optimization scenarios, achieving the lowest mean value of the predefined management objective, with a 3.9% reduction in the fixed-weight comprehensive score compared to the state-of-the-art optimizer, and maintains a perfect 100% feasible policy rate across five independent random seeds under unified high-fidelity safety rechecking. Third, the designed ablation study rigorously pinpoints the three dominant sources of HCMBO's performance advantage: the joint optimization of direction and rate control variables, the deliberate avoidance of myopic short-horizon hard screening that discards potentially high-quality solutions, and the proposed structured direction-wise rate search mechanism that reduces redundant exploration in the high-dimensional decision space.

The real-world transfer experiment inspired by the Shanghai Bund scenario further demonstrates the strong practical generalization capability of the proposed framework. Derived purely from the HCMBO optimizer, the optimized crowd management policy reorganizes pedestrian flow in a high-fidelity spatial abstraction of the hotspot, reducing the overall management objective by 26.9% by dynamically steering excess crowds away from the severely overloaded terminal channel and rebalancing pedestrian load across the entire middle-channel group. Collectively, these quantitative results establish the proposed framework as an effective, interpretable, and deployable foundation for proactive, optimization-driven crowd management in complex public pedestrian spaces.

TABLE IX
PARAMETER CORRESPONDENCE BETWEEN THE TRANSFER EXPERIMENT AND A REAL-SCALE BUND OPERATING CONDITION

Parameter	Transfer experiment	Real-scale condition	Effect on cost
Spatial extent	$\approx 894 \text{ m} \times 391 \text{ m}$ (georeferenced)	identical (covers the waterfront core area)	unchanged
Grid resolution	512×224 , $\Delta x \approx 1.75 \text{ m}$	identical	sole spatial cost factor; unchanged
Time step	$\Delta t \approx 0.44 \text{ s}$ (CFL-induced)	identical (independent of crowd size)	unchanged
Instantaneous population	peak $\approx 1.5 \times 10^3$ ped	$5 \times 10^4 - 7 \times 10^4$ ped	none (density values only)
Peak local density	$\approx 4.0 \text{ ped/m}^2$ ($= \rho_{\max}$)	same saturated regime	none; regime already covered
Total entrance inflow	$\approx 3.35 \text{ ped/s}$ (single main entrance)	$\approx 20-30 \text{ ped/s}$ (multiple entrances, field-calibrated)	none (boundary condition)
Simulation horizon	$\approx 698 \text{ s}$ (1600 steps)	1–2 h peak period ($\approx 8 \times 10^3 - 1.6 \times 10^4$ steps)	total cost grows linearly with steps

This study also acknowledges several clear limitations that point to promising future research directions. First, all current experiments adopt simplified 2D geometric representations of the hotspot areas, which do not fully capture the 3D topological constraints of multi-level pedestrian facilities such as overpasses, underground passages, and connected commercial podiums. Second, the current multi-stage route behavior model does not incorporate real-time pedestrian feedback dynamics, such as adaptive route switching triggered by visible crowd congestion or real-time mobile navigation guidance. Third, the optimization framework is currently designed for centralized offline planning, and further extensions are required to support real-time rolling horizon optimization under dynamic, uncertain crowd arrival patterns. These limitations will be systematically addressed in our subsequent work to further improve the deployability of the proposed system in large-scale real-world tourism hotspot scenarios.

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